PRIME SCHOOL

Science

Year 7

Cambridge Lower Secondary
Student Manual

PRIME BOOKS · SCIENCE YEAR 7 · STUDENT BOOK

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Diagrams and facts. All diagrams are original to this edition. Scientific facts are drawn from real, credited sources listed in the Sources & references section.

British English edition. Typeset in Source Sans, with display set in a modern serif.

CONTENTS

What is inside this book

INTRODUCTION

How this book works

Science is not a list of facts to memorise. It is a way of asking questions about the world and finding answers you can trust. This book is built around that idea. You will learn the big ideas of biology, chemistry and physics, but you will also learn to think and work like a scientist: to observe carefully, to measure honestly, to plan fair tests and to draw conclusions from evidence.

In this first unit you will look closer than you ever have before, past what your eyes can see, into the tiny world of the cell. You will discover that every living thing on Earth, from the cork oaks of the Alentejo to you, is built from the same astonishing units.

What makes this book different

- **You do science, not just read it.** Every unit has investigations where

you use real equipment and real methods.

- **Thinking is taught explicitly.** A dedicated strand shows you how

scientists plan, measure, analyse and conclude.

- **Science connects to life.** You will see how the ideas in this book are

used in the world beyond the laboratory.

A note to teachers and parents

This book follows the spirit of an international lower-secondary science programme, interpreted afresh for Prime School, across three strands: Science, wherever an investigation requires it.

HOW TO USE THIS BOOK

Reading the page

Each topic follows the same rhythm.

PANEL	WHAT IT DOES
In this topic, you will	The goals for the topic.
Getting started	A quick warm-up to wake up what you know.
Diagram	A labelled scientific drawing to study.
Questions	Quick checks as you read.
Think like a scientist	A task that builds scientific thinking.
Did you know?	A true and surprising scientific fact.
Summary checklist	"I can" statements to check your progress.

How to work

Read with curiosity. When the book asks a question, try to answer it before you read on. In investigations, follow the method exactly and record what you actually observe, not what you expected to see. Honest results are the heart of good science.

GETTING SET UP

Before Unit 1

For this unit you will mostly need your eyes, your curiosity and a pencil. For the investigations, your teacher will provide microscopes, slides and stains.

Your scientist's toolkit

- A sharp pencil and a ruler, for careful labelled diagrams.
- A notebook for observations and results.
- Care. Microscopes and stains must be used exactly as instructed.

A quick self-check

1. Name three things that are living and three that are not. How do you know?
2. What is a microscope used for?
3. Roughly how big do you think a single cell is?

The answers are exactly what Unit 1 will make clear. When you can describe what a cell is and why it matters, you are ready to begin.

UNIT 1

1

Cells

The building blocks of every living thing

IN THIS UNIT, YOU WILL

- ✓ Explain what a cell is and how the microscope revealed them.
- ✓ Identify the parts of animal and plant cells and their functions.
- ✓ Explain how specialised cells are adapted to their jobs.
- ✓ Describe how cells build tissues, organs and organisms.

✦ GETTING STARTED

Everything alive is built from cells, but almost nobody saw one until 350 years ago. In pairs, discuss:

- why do you think cells stayed unknown for so long?
- what would you need in order to see one?
- if every living thing is made of cells, what might a tree and a fish have in common?

Cells: the units of life

◎ **IN THIS TOPIC YOU WILL LEARN**

- what a cell is, and why every living thing is made of them
- how the microscope made cells visible for the first time
- how to calculate the magnification of an image

Every living thing is built from cells

A **cell** is the smallest unit of a living thing that can carry out all the processes of life. Every organism you have ever seen is built from cells: the grass on a football pitch, the mould on old bread, the fish in the Tejo, and you. This idea is so important in biology that it has a name, the **cell theory**, and it makes three claims:

1. All living organisms are made of one or more cells.

2. The cell is the basic unit of structure and function in living things.

3. All cells come from cells that already exist.

That third point matters. Cells do not appear from nothing. Every cell in your body was produced when an earlier cell divided into two, and you can follow that chain back, division by division, to a single fertilised egg cell.

Some organisms are **unicellular**, meaning the whole organism is one cell. An *Amoeba* in pond water is a single cell that feeds, moves, senses its surroundings, grows and reproduces. Everything an organism must do, that one cell does. Others are **multicellular**. An adult human is built from roughly thirty trillion cells, and those cells are not all the same: they are specialised into more than two hundred different types.

Cells are extremely small

Most cells are far too small to see. A typical animal cell is about 20 **micrometres** across. A micrometre is one thousandth of a millimetre, so you could line up fifty such cells across a single millimetre on your ruler. This is why cells stayed unknown for almost all of human history: nobody could see them.

There are striking exceptions. The yolk of a hen's egg is a single cell, and some nerve cells run from the base of your spine to your toes, making them nearly a metre long, though still microscopically thin.

The microscope changed everything

In 1665 the English scientist **Robert Hooke** examined a thin slice of cork with a microscope he had built himself. He saw a honeycomb of tiny empty boxes and named them *cells*, from the Latin *cella*, meaning a small room, because they reminded him of the rooms in a monastery. What Hooke actually saw were the empty cell walls of dead plant tissue, but the name stayed.

A few years later, in the Netherlands, **Antonie van Leeuwenhoek** ground much better lenses and became the first person to see living cells: bacteria, single-celled organisms in pond water, and blood cells. He called them *animalcules*, little animals.

✧ DID YOU KNOW?

Van Leeuwenhoek's best lenses magnified about 270 times, which is close to what a good school microscope achieves today. He never told anyone how he made them, and his method was not rediscovered for over two hundred years.

Working out magnification

A microscope makes an object appear larger. How much larger is the **magnification**, and you calculate it like this:

magnification = size of image ÷ actual size of object

Both measurements must be in the same unit before you divide.

■ WORKED EXAMPLE 1.1

A cell is 0.05 mm across. In a photograph it measures 30 mm across. What is the magnification?

Convert to the same unit: the actual size is 0.05 mm and the image is 30 mm.

magnification = 30 ÷ 0.05 = **600**

The image is 600 times larger than the cell, written ×600.

■ WORKED EXAMPLE 1.2

A photograph is labelled ×400. The image of a cell measures 24 mm. How wide is the real cell?

Rearrange the formula: actual size = size of image ÷ magnification

actual size = 24 ÷ 400 = 0.06 mm, which is **60 micrometres**.

Questions

RECALL

1. Write down the three statements of the cell theory. [3]

2. What does the word *unicellular* mean? Give one example. [2]

3. Who first used the word *cell*, in which year, and what was he looking at? [3]

4. How many micrometres are there in one millimetre? [1]

APPLY

5. A cell is 0.02 mm wide. Its image is 40 mm wide. Calculate the magnification. [2]

6. An image is taken at ×250 and measures 35 mm. Calculate the actual size of the object in millimetres, then convert your answer to micrometres. [3]

7. A pupil says a hen's egg cannot be a cell because it is far too big. Explain why the pupil is wrong. [2]

EXPLAIN

8. Explain why the invention of the microscope was necessary before the cell theory could be proposed. [3]

9. Hooke saw only empty boxes in his cork. Suggest why he saw no living contents inside them. [2]

ANALYSE

10. Van Leeuwenhoek reported seeing tiny moving organisms in a drop of water. Many scientists did not believe him. Suggest two reasons why other scientists could not confirm his observations at the time. [2]

Animal cells and plant cells

IN THIS TOPIC YOU WILL LEARN

- the parts of an animal cell and what each one does
- the three extra structures found only in plant cells
- how to tell a plant cell from an animal cell under a microscope

Inside an animal cell

Every animal cell is surrounded by a **cell membrane**. This is a thin, flexible layer that holds the cell together and controls what passes in and out. It is described as *partially permeable*: oxygen and dissolved food are allowed through, while other substances are kept out. Nothing enters or leaves a cell without crossing this membrane.

Inside, the cell is filled with **cytoplasm**, a jelly-like fluid that is mostly water. This is not simply a filler. The cytoplasm is where most of the cell's chemical reactions happen, and the structures of the cell are suspended in it.

The **nucleus** is usually the largest structure in an animal cell. It controls everything the cell does, because it contains the cell's **DNA**, the coded instructions for building and running the whole organism. When a cell divides, the nucleus makes a copy of those instructions so each new cell receives a complete set.

Scattered through the cytoplasm are **mitochondria** (one of them is a *mitochondrion*). These release energy from food in the process of respiration. Cells that need a great deal of energy contain far more of them: a muscle cell may hold a few thousand mitochondria, while a skin cell holds far fewer.

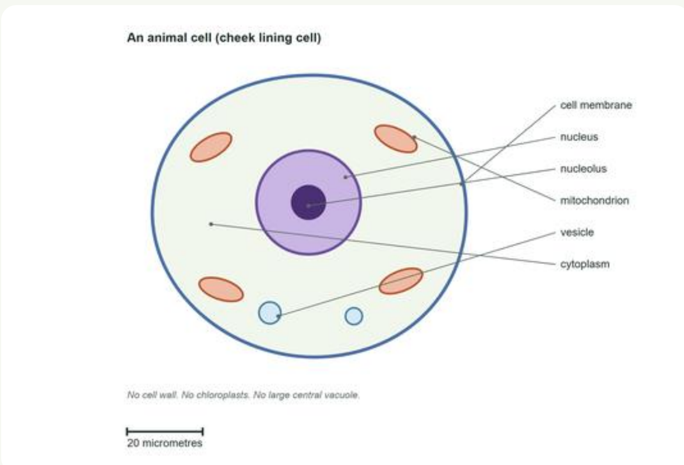

Figure 1.1 An animal cell. Notice the irregular outline and the absence of a cell wall.

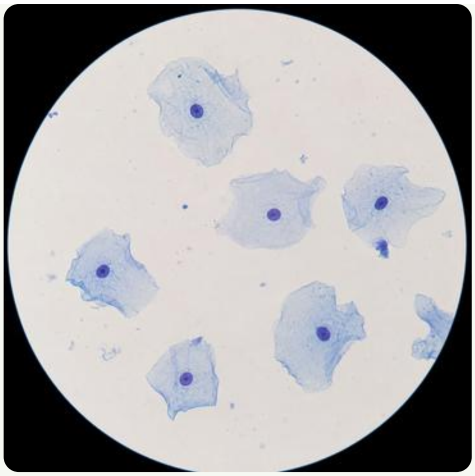

Figure 1.2 Cheek lining cells seen under a light microscope, stained with methylene blue. Illustrative micrograph-style rendering.

Inside a plant cell

A plant cell contains everything an animal cell contains: membrane, cytoplasm, nucleus and mitochondria. It then adds three structures of its own.

Outside the cell membrane lies the **cell wall**, made of a tough fibre called cellulose. Be careful here: the wall is *outside* the membrane, and the two are different things. The wall is fully permeable, so it does not control what enters, but it gives the cell strength and a definite shape. This is why plant cells look like neat boxes while animal cells look rounded and irregular, and it is why a plant can stand upright without a skeleton.

Plant cells that are exposed to light contain **chloroplasts**. These small lens-shaped structures hold the green pigment **chlorophyll**, which absorbs light energy so the plant can make glucose by photosynthesis. Not every plant cell has them: root cells are underground in the dark, so they have none.

A mature plant cell also has one large **central vacuole**, a space filled with cell sap that can occupy more than half the cell's volume. When it is full it presses outwards against the cell wall and keeps the cell firm. When a plant is short of water the vacuoles shrink, the cells lose their firmness, and the plant wilts.

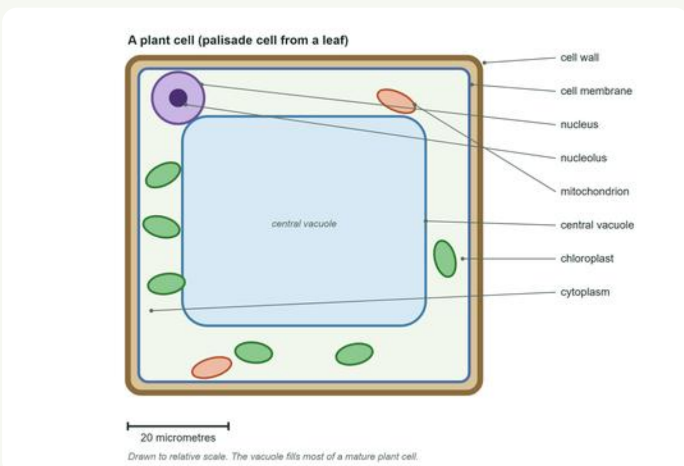

Figure 1.3 A plant cell. The wall lies outside the membrane and the vacuole dominates the cell.

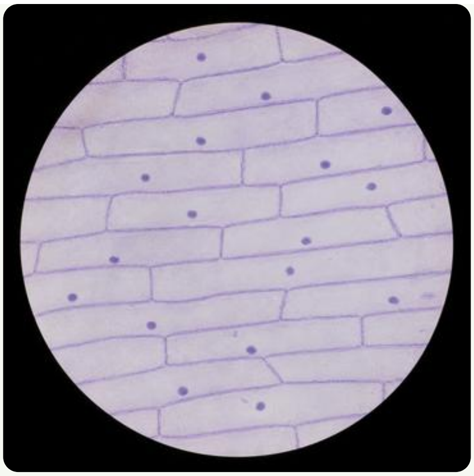

Figure 1.4 Onion epidermis under a light microscope. The brick-like walls are clearly visible. Illustrative micrograph-style rendering.

Comparing the two

STRUCTURE	ANIMAL CELL	PLANT CELL	WHAT IT DOES
cell membrane	yes	yes	controls what enters and leaves
cytoplasm	yes	yes	where reactions happen
nucleus	yes	yes	controls the cell, holds DNA
mitochondria	yes	yes	release energy in respiration
cell wall	no	yes	gives strength and shape
chloroplasts	no	in cells exposed to light	absorb light for photosynthesis
large vacuole	no	yes	stores sap, keeps the cell firm

✧ DID YOU KNOW?

Animal cells do contain small vacuoles, but they are tiny and there may be several. Only plant cells have the single large permanent vacuole.

Questions

RECALL

1. Name the four structures found in both plant and animal cells. [4]

2. Name the three structures found only in plant cells. [3]

3. What is the cell wall made of? [1]

4. Which structure contains the cell's DNA? [1]

5. What is the green pigment inside a chloroplast called? [1]

APPLY

6. A cell is examined and found to have a nucleus, a cell wall and a large vacuole, but no chloroplasts. Is it a plant cell or an animal cell? Give a reason, and suggest where in the plant it came from. [3]

7. Muscle cells contain many more mitochondria than skin cells. Explain why. [2]

8. Copy the table above and add one further row for a structure of your own choosing. [2]

EXPLAIN

9. Explain the difference between the cell wall and the cell membrane, referring to their position, what they are made of, and what each one does. [4]

10. Explain why a plant wilts when it has not been watered, using the words *vacuole*, *cell wall* and *firm*. [3]

11. Root cells have no chloroplasts. Explain why this does not prevent the root from getting the energy it needs. [3]

ANALYSE

12. A pupil looks at an unknown cell and says: "It has no chloroplasts, so it must be an animal cell." Explain why this conclusion is unsafe, and state what she should look for instead. [3]

Specialised cells

◎ **IN THIS TOPIC YOU WILL LEARN**

- what it means for a cell to be specialised
- how the shape of a cell is suited to the job it does
- several examples of specialised plant and animal cells

Why cells become different

A unicellular organism such as an *Amoeba* must do everything with one cell. A multicellular organism can afford to divide the work up, and that is exactly what happens. As an organism grows, its cells become **specialised**, developing a particular shape and contents that suit one job. This process is called **differentiation**.

The rule to remember is that **structure fits function**. If you are shown an unfamiliar cell, look at its shape and ask what that shape would be good for. The answer is usually the job it does.

Root hair cells: absorbing water

A root hair cell has one long, narrow extension that pushes out between the soil particles. That extension enormously increases the cell's **surface area**, and because water enters across the surface, a larger surface means faster absorption. A single rye plant can carry billions of root hairs; laid end to end they would stretch for hundreds of kilometres.

These cells have no chloroplasts, because there is no light underground.

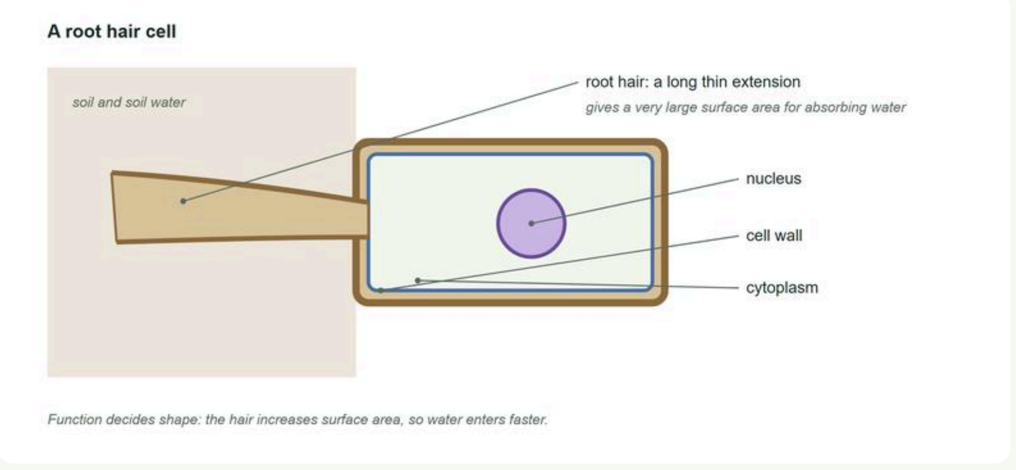

Figure 1.5 A root hair cell. The long extension gives a very large surface area for absorbing water.

Red blood cells: carrying oxygen

A red blood cell is shaped like a disc that has been pressed in on both sides, a shape called **biconcave**. This gives it a larger surface area than a sphere of the same volume, so oxygen is absorbed and released more quickly.

More unusually, a mature red blood cell has **no nucleus**. Losing the nucleus frees up space for more **haemoglobin**, the red protein that binds oxygen. The cost is that the cell cannot divide or repair itself, so it survives only about 120 days before being replaced.

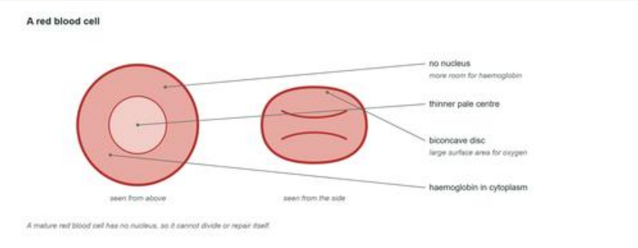

Figure 1.6 A red blood cell from above and from the side.

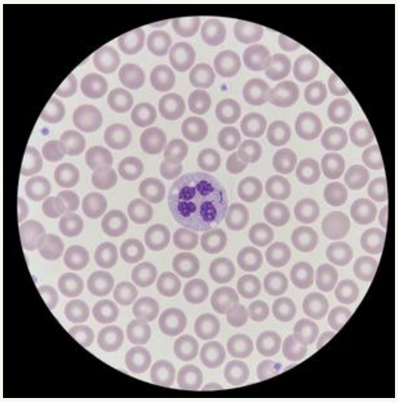

Figure 1.7 A blood smear. The large cell with the dark lobed nucleus is a white blood cell. Illustrative rendering.

Nerve cells: carrying messages

A nerve cell, or **neurone**, is extremely long and thin. Its job is to carry electrical impulses from one part of the body to another, and length is exactly what that requires. The longest neurones in your body run from the base of the spine to the toes: a single cell almost a metre long.

At each end the cell branches into fine extensions, so one neurone can pass its message on to many others.

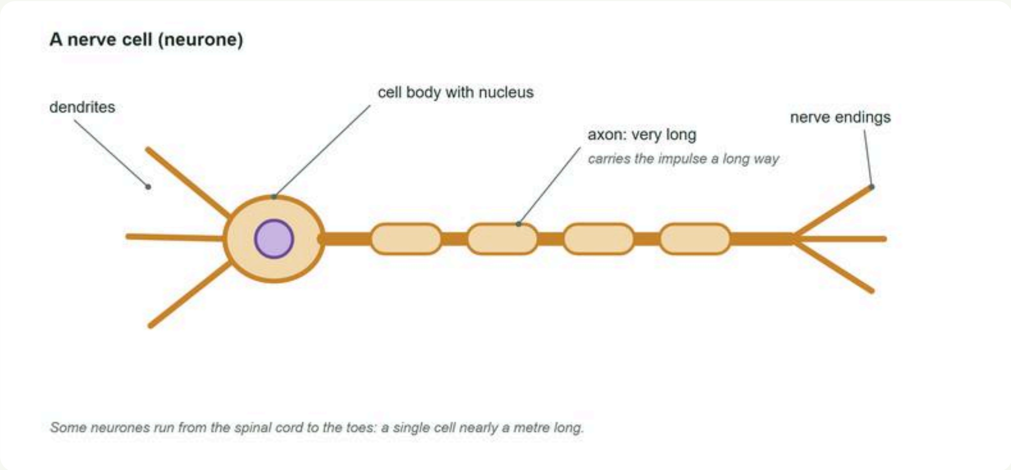

Figure 1.8 A nerve cell. The axon may be very long indeed.

Palisade cells: making food

Palisade cells sit in a tightly packed layer just below the upper surface of a leaf, exactly where the light is strongest. They are tall and column-shaped, so many can be fitted side by side, and each is packed with chloroplasts. This is where most photosynthesis in a plant takes place.

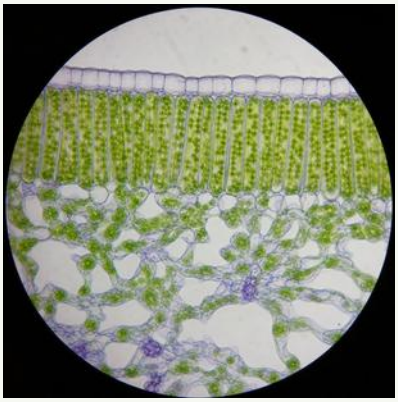

Figure 1.9 A section through a leaf: tall palisade cells packed with chloroplasts. Illustrative rendering.

Figure 1.9b A nerve cell under the microscope. Illustrative rendering.

Comparing specialised cells

CELL	SPECIAL FEATURE	WHY THE FEATURE HELPS
root hair	long thin extension	large surface area to absorb water
red blood	biconcave, no nucleus	large surface area, more room for haemoglobin
nerve	very long axon	carries impulses a long distance
palisade	tall, many chloroplasts	absorbs the most light for photosynthesis
ciliated	tiny beating hairs	sweeps mucus and dust out of the airways

✧ **DID YOU KNOW?**

Your body makes about two million red blood cells every second, simply to replace the ones that wear out.

Questions

RECALL

1. What does *specialised* mean when describing a cell? [1]

2. Name the process by which cells become specialised. [1]

3. Give two features of a red blood cell that help it carry oxygen. [2]

4. Where in a leaf are palisade cells found? [1]

APPLY

5. For each cell, name one feature and say what it is for:

a. root hair cell b. nerve cell c. palisade cell [6]

6. A cell is long and thin with many mitochondria and fine hairs along one edge. Suggest what job it does, and give a reason. [2]

7. Root hair cells contain no chloroplasts. Explain why this is not a problem for the plant. [2]

EXPLAIN

8. Explain how the biconcave shape of a red blood cell helps it do its job. [3]

9. Explain why a mature red blood cell cannot repair itself, and state one consequence. [3]

10. Explain why palisade cells are found at the top of a leaf rather than the bottom. [3]

ANALYSE

11. Both root hair cells and red blood cells have a large surface area, but for different reasons. Compare the two, saying what each absorbs. [4]

12. Suggest why an organism made of only one cell cannot have specialised cells, and explain one disadvantage this creates. [3]

From cells to organisms

◎ **IN THIS TOPIC YOU WILL LEARN**

- how cells are organised into tissues, organs and organ systems
- examples of each level in both animals and plants
- why this organisation makes a large organism possible

Five levels of organisation

A multicellular organism is built in layers, each one made from the level below.

Cell → **tissue** → **organ** → **organ system** → **organism**

A **tissue** is a group of similar cells working together on the same job. Muscle tissue is made of muscle cells that all contract. Xylem tissue in a plant is made of cells that all carry water.

An **organ** is several different tissues working together as a unit. The stomach is an organ: it contains muscle tissue to churn food, glandular tissue to make digestive juices, and epithelial tissue lining the inside. No single tissue could digest food alone.

An **organ system** is a group of organs that together carry out one major function. The digestive system includes the mouth, oesophagus, stomach, intestines, liver and pancreas.

Finally, all the organ systems working together make the whole **organism**.

Examples in animals

LEVEL	EXAMPLE
cell	a muscle cell
tissue	muscle tissue
organ	the heart
organ system	the circulatory system
organism	a human being

The human body has about a dozen organ systems, including the circulatory, digestive, respiratory, nervous and skeletal systems. They are not independent: the circulatory system delivers the oxygen the respiratory system collects.

Examples in plants

Plants are organised in exactly the same way.

LEVEL	EXAMPLE
cell	a palisade cell
tissue	palisade tissue
organ	a leaf
organ system	the shoot system (stem, leaves, flowers)
organism	an olive tree

A leaf really is an organ: it contains palisade tissue for photosynthesis, spongy tissue with air spaces for gas exchange, xylem to bring water and phloem to carry sugars away.

Figure 1.10 Root hair cells absorb water, which xylem tissue then carries up through the whole plant. Illustrative micrograph-style rendering.

Why organisation matters

A single cell can only survive at a small size, because substances enter and leave across its surface. As anything grows, its volume increases faster than its surface area, so a very large single cell could not absorb enough oxygen or food to supply its middle.

Being multicellular removes that limit. Specialised cells form tissues, tissues form organs, and organs can serve the whole body: lungs collect oxygen for every cell, and blood delivers it. That is what makes a large, complex organism possible.

✧ DID YOU KNOW?

The largest organ in your body is your skin. In an adult it covers about two square metres and weighs around four kilograms.

Questions

RECALL

1. Write out the five levels of organisation in order, starting with the cell. [2]

2. Define the word *tissue*. [1]

3. Define the word *organ*. [1]

4. Name three organs of the human digestive system. [3]

APPLY

5. Put each of these at the correct level: the heart, a palisade cell, xylem, the nervous system, an olive tree. [5]

6. Give one example of a tissue and one example of an organ in a plant. [2]

7. Name the organ system that contains the lungs, and state its main job. [2]

EXPLAIN

8. Explain why the stomach is described as an organ rather than a tissue. [3]

9. Explain why a leaf is an organ. Name two tissues it contains and give the job of each. [4]

10. Explain why organ systems must work together, using the circulatory and respiratory systems as your example. [3]

ANALYSE

11. Explain why a single cell cannot grow to the size of a football, referring to surface area and volume. [4]

12. Suggest two advantages a multicellular organism has over a unicellular one, and one disadvantage. [3]

Check what you know

◎ ABOUT THIS TEST

There are 14 questions and 60 marks. Marks are shown in brackets. Answer every question in full sentences unless the question tells you otherwise.

1. Copy and complete these sentences. [4]

a. The smallest unit of a living thing that carries out all life processes is the ....

 a. A group of similar cells working together is a ....
 b. The structure that controls what enters and leaves a cell is the ....
 c. The green pigment that absorbs light is called ....

2. State the three parts of the cell theory. [3]

3. The diagram in Figure 1.3 shows a plant cell.

a. Name the structure that lies outside the cell membrane. [1]

 a. Name the structure that fills most of the cell. [1]
 b. Name two structures present in this cell that would **not** be found in an animal cell, other than your answer to part a. [2]

4. Copy and complete the table with a tick or a cross. [6]

STRUCTURE	ANIMAL CELL	PLANT CELL
nucleus		
cell wall		
chloroplast		

5. A cell measures 0.04 mm across. In a photograph it measures 32 mm across. Calculate the magnification. Show your working. [3]

6. An image is taken at ×500. The image of a cell measures 45 mm.

a. Calculate the actual width of the cell in millimetres. [2]

 a. Convert your answer to micrometres. [1]

7. Explain the difference between the cell wall and the cell membrane. Refer to their position, what they are made of, and what each does. [4]

8. A pupil examines an unknown cell. It has a nucleus and a cell wall but no chloroplasts.

a. Is it a plant cell or an animal cell? Give a reason. [2]

   a. Suggest where in the plant the cell came from, and explain your answer. [2]

9. Describe **two** ways a red blood cell is adapted to carry oxygen, and explain how each adaptation helps. [4]

10. Root hair cells have a long, thin extension.

a. What does this shape increase? [1]

   a. Explain how that helps the plant. [2]

   b. Explain why root hair cells contain no chloroplasts. [2]

11. Write the five levels of organisation in order, from smallest to largest, and give one animal example of each. [5]

12. Explain why a leaf is described as an organ. Name two tissues it contains and state the job of each. [4]

13. Muscle cells contain many more mitochondria than skin cells.

a. What do mitochondria do? [1]

   a. Explain why muscle cells need more of them. [2]

14. A pupil says: "A plant wilts because its cell walls collapse."

a. Explain why this statement is wrong. [2]

   a. Give the correct explanation, using the words *vacuole*, *water* and *firm*. [3]

◎ **MARKS**

Total: 60 marks.

GLOSSARY

Unit 1

cell: the smallest unit of a living thing that can carry out all the processes of life.

cell membrane: the thin partially permeable layer that controls what enters and leaves a cell.

cell theory: the idea that all living things are made of cells, that the cell is the basic unit of life, and that all cells come from existing cells.

cell wall: a tough layer of cellulose outside the membrane of a plant cell, giving strength and shape.

chlorophyll: the green pigment inside a chloroplast that absorbs light.

chloroplast: the structure in a plant cell where photosynthesis happens.

cytoplasm: the jelly-like fluid where most of a cell's chemical reactions take place.

differentiation: the process by which a cell becomes specialised.

DNA: the molecule carrying the coded instructions for building an organism.

haemoglobin: the red protein in red blood cells that binds oxygen.

magnification: how many times larger an image is than the real object.

micrometre: one thousandth of a millimetre.

mitochondrion: the structure that releases energy from food in respiration.

multicellular: made of many cells.

neurone: a nerve cell, adapted to carry electrical impulses.

nucleus: the structure that controls the cell and contains its DNA.

organ: several different tissues working together, such as the heart.

organ system: a group of organs carrying out one major function.

specialised: having a shape and contents suited to one particular job.

tissue: a group of similar cells working together.

unicellular: made of a single cell.

vacuole: a space filled with cell sap; in a mature plant cell it is large and permanent, and keeps the cell firm.

SOURCES & REFERENCES

Where this material came from

The scientific content of this unit follows the course textbook used at Prime School. All diagrams in this unit were drawn for this book, to scale, and the micrograph-style plates are illustrative renderings prepared for teaching, not photographs of specimens.

§ UNIT 1 · CELLS

1. **Course textbook**, Lower Secondary Science Stage 7, Unit 1 "Cells", the unit on cell structure, specialised cells, and organisation from cells to organisms. This is the source followed throughout this unit.
2. **Robert Hooke**, *Micrographia*, 1665. The observation of cork and the first printed use of the word *cell*.
3. **Antonie van Leeuwenhoek**, letters to the Royal Society, from 1674. The first recorded observations of living single-celled organisms.
4. **Prime Books studio**, original scale diagrams and micrograph-style plates prepared for this unit, 2026.

Figures 1.1, 1.3, 1.5, 1.6 and 1.8 are original scale diagrams. Figures 1.2, 1.4, 1.7, 1.9 and 1.10 are illustrative micrograph-style renderings.

PRIME BOOKS

Science

Year 7 · Cambridge Lower Secondary · Student Manual

Ask questions. Trust evidence.

Year 7 science combines clear explanations with hands-on investigation across biology, chemistry and physics, building the habits of working scientifically.

INSIDE THIS BOOK

- Biology, chemistry and physics strands
- Practical investigations throughout
- Diagrams, data and vocabulary support
- End-of-unit checkpoints
- Full glossary and answer key

Prime Books · Science

Ages 11–12 · Lower Secondary

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